WEEK 1 · LECTURE L2 · UNIT I

Why HR Is Going Digital

Industry 4.0, artificial intelligence and hybrid work; the difference between digitizing and transforming; and what the three together are doing to the HR function and to the nature of work.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Explain the three forces reshaping HR (Industry 4.0, artificial intelligence, hybrid and platform work) and Schwab’s case that this revolution is new in kind.
2. Distinguish digitization, digitalization and transformation (Verhoef et al.), and explain stalls with Vial’s building blocks.
3. Apply Stone et al., Zuboff, Autor, Levy and Murnane, and Orlikowski to an HR change, and state the limits of each.
4. Argue why imported digital HR frameworks need adapting to Indian organizational conditions.

ONE SOURCE

The same theories as the in class lecture and the course hub. This edition leaves out the simulation data used in class, which the end term examination does not use. Course outcomes: CLO 1 and CLO 5.

OPENING QUESTION

What actually changes when HR goes digital?

Who gets faster

And who quietly gets slower, because the new system was not designed around their device, language or job?

What judgment is encoded

Which decision moves from a person to a rule, and who can still see that rule and question it?

Whose data moves

What does the organization now know about each employee that it did not know before, and who controls it?

COURSE FRAMING

Hold these three questions open. Every theory in this lecture answers at least one of them.

COMMIT BEFORE THEORY

Is this digital transformation?

An organization scans all its paper leave forms into PDFs and emails them to managers. Is this digital transformation?

A

Yes: the process is now digital

B

No: it digitizes a record but leaves the process, roles and decisions unchanged

C

Yes, if it saves paper

D

Only if it uses AI

EXAM LENS

The answer is B. Commit to an answer before you read on; the theories that follow explain why.

THE DRIVERS

Three forces, arriving together

I · Industry 4.0

Cyber physical systems, the internet of things, cloud and connectivity collapse the distance between the shop floor and the data layer. Work becomes measurable in real time.

II · Artificial intelligence

Prediction becomes cheap. Screening, matching, ranking and forecasting move from human judgment to model output: fast and scalable, but harder to explain.

III · Hybrid and platform work

Where work happens, and under what contract, both come loose from the office and the payroll. The workforce stops being one population.

COURSE FRAMING

None of these is an HR initiative, yet all three land on HR. That is the core argument for why the HR function must change shape.

FORCE I · SCHWAB (2016)

Four industrial revolutions

Revolution	When	Trigger	What it brought
First	About 1760 to 1840	Railroads and the steam engine	Mechanical production
Second	Late 19th to early 20th century	Electricity and the assembly line	Mass production
Third	From the 1960s	Semiconductors, mainframes, PCs, the internet	The computer or digital revolution
Fourth	Now	Builds on the digital revolution	Fuses the physical, digital and biological worlds

EXAM LENS

Use Schwab’s dates, as above. Other timelines (for example 1784, 1870, 1969) come from other sources; if you name Schwab, cite his.

Dates and triggers as given in Schwab (2016), Introduction and Part I.

What each revolution did to work and to people management

Revolution	What it did to work	What it meant for people management
First	Steam mechanization moved work from homes to factories	HR did not exist; labor was hired at the factory gate
Second	Electricity and assembly lines created specialized, repetitive tasks	Personnel administration appears: records, payroll, discipline
Third	Electronics and IT introduced knowledge work	HRM professionalizes; systems of record (the HRIS) arrive
Fourth	Connectivity and AI reshape the relationship between people and machines	Every process emits data; HR must decide with it and answer for it

EXAM LENS

For HR the question is not which tool to buy, but which systems of work are being redrawn.

Column 2 from the course's Unit 1 teaching note; column 3 is Course framing. Dates as in Schwab on the previous page.

FORCE I · SCHWAB (2016)

Why the fourth is a new revolution

1 · Velocity

Change is exponential, not linear: each technology accelerates others.

2 · Breadth and depth

Several technologies combine and change what is done and who does it, not only how.

3 · Systems impact

Whole systems shift at once: companies, industries, countries and society.

For HR, the point is that the function must redesign work, not only automate its own administration.

EXAM LENS

A strong answer uses all three criteria, not just “it is faster”.

FORCE I · SCHWAB (2016)

Effects on business and on people

Four effects on business

Customer expectations shift · products are enhanced by data · new partnerships form · operating models become digital, including platforms that match supply and demand without employing the supply: the economic origin of the gig workforce (Week 10)

Implications for HR

Talent more than capital becomes the critical factor of production; a shortage of skilled workers, not of capital, is likely to limit growth · skills go out of date faster, so learning becomes continuous · work fragments into tasks · gains flow to those who can work with the new technologies

Critique

Critics argue the label overstates novelty and that much of the change continues the computing revolution. The view is global and high bandwidth; in India the change arrives unevenly, reaching desk staff before phone only frontline staff. Ask who the change reaches.

EXAM LENS

Common mistake: treating Industry 4.0 as a technology list. A firm can buy every listed technology and remain a third revolution organization.

The nine pillars, and what they mean for HR

1. **Big data and analytics**
   Evidence based workforce decisions
2. **Autonomous robots and automation**
   Process automation in screening, enrollment, payroll
3. **Simulation and digital twins**
   Workforce scenario models; VR training
4. **Horizontal and vertical integration**
   Connected systems across HR and the business
5. **Industrial internet of things**
   Wearables and sensors, with surveillance risk
6. **Cybersecurity**
   Protecting employee data
7. **Cloud computing**
   HR systems as services
8. **Additive manufacturing**
   A metaphor for personalization at scale
9. **Augmented and virtual reality**
   Immersive learning

HR skills Industry 4.0 demands: data literacy, technology acumen, business acumen, change management expertise, ethical reasoning.

The nine pillars are BCG's (Rüßmann et al., 2015). HR meanings from the course's Unit 1 teaching note; the labels for pillars 4, 6 and 7 are the course's. The term Industry 4.0 began in German manufacturing policy in 2011 (Kagermann et al., 2013).

SCULLY-RUSS AND TORRACO (2020)

What technology does to work itself

Their review traces why work is changing: technology, finance, worker behavior and institutional shifts (pp. 73 to 77, read fully in Week 2). Four shifts follow that HR meets first:

Tasks split from jobs

A job becomes a shifting portfolio of tasks, some of which move to machines each year.

Skills date faster

Learning becomes continuous infrastructure, not a one time event.

Work emits data

Logins, scans, routes and clicks are recorded whether or not anyone designed a measurement system.

The firm loses its edges

Employees, contractors, apprentices and platform workers work side by side under different contracts.

EXAM LENS

Link each shift to the HR process it disrupts first: job design, learning, privacy, workforce planning.

The four shifts are Course framing, drawn from the review.

FORCES II AND III

What AI and hybrid work ask of HR

AI automates

Routine cognitive work: policy queries, CV screening, scheduling.

AI augments

Human judgment: predicting attrition, recommending learning.

AI creates new risks

Biased decisions, opaque reasoning and new duties under data protection law (Week 9).

HR is therefore both a user of AI and the function accountable for its effects on people.

HYBRID WORK

Since 2020 many desk roles have moved to a mix of office and remote work. HR must deliver services, learning and engagement to people it rarely sees, measure performance by output rather than presence, and protect fairness between those who can work remotely and those who cannot (Week 10).

Where algorithms already sit in the employee lifecycle

Stage	Typical algorithmic use	What improves	What can get worse
Attract and screen	CV ranking, keyword filters, chatbot prescreening	Speed and consistency at volume	Past bias is learned and scaled
Onboard and learn	Document verification, content recommendation	Administrative load falls	Reasons for decisions become invisible
Perform and reward	Goal cascades, calibration, pay modeling	Patterns no manager could see unaided	Accountability diffuses: “the model decided”
Predict and retain	Flight risk scores, succession, headcount forecasts	Earlier intervention	Prediction treated as fact

Course framing.

FORCE III · GRATTON (2021)

Hybrid work, and the equity trap

Two axes

Place (office to anywhere) and time (synchronous, fixed hours, to asynchronous).

Four productivity drivers

For each job, ask what it needs for energy, focus, coordination and cooperation.

FOUR PERSPECTIVES FOR DESIGN

Jobs and tasks

Which tasks need energy, focus, coordination or cooperation?

Employee preferences

Autonomy over place and time is now a retention variable, not a perk *

Projects and workflows

Redesign the workflow, or hybrid simply degrades it *

Inclusion and fairness

Who is eligible for flexibility, and who is structurally excluded? *

COURSE FRAMING

The trap: a store associate, a factory operator or a delivery partner cannot work from home. If flexibility is designed only for laptop users, hybrid work divides the workforce in two, and HR built the wall. Week 10 develops this.

Axes, perspectives and drivers from Gratton (2021). * Course framing.

CONTEXTUALIZING THE FORCES

The Indian layer: why imported frameworks slip

The divide is inside the firm

Corporate offices are often well digitized while store floors, factories and warehouses remain manual. Vernacular languages, shared devices and thin bandwidth are common.

Platform work is scaling fast

NITI Aayog (2022) estimated about 7.7 million gig workers in 2020 to 21, projected to reach about 23.5 million by 2029 to 30.

The law has arrived

The Digital Personal Data Protection Act, 2023 governs employee data and consent; the Code on Social Security, 2020 and the Rajasthan Act, 2023 reach platform work.

Public digital rails

India Stack (Aadhaar based verification, DigiLocker, UPI) enables onboarding and payment designs most global HR vendors never built for, with privacy tradeoffs of their own.

VERHOEF ET AL. (2021)

Three depths of change

1 · Digitization

Analog information becomes digital. The process is untouched.

A paper personnel file becomes a PDF

2 · Digitalization

A process moves onto a system but keeps its old logic.

The annual engagement survey moves online, still once a year

3 · Digital transformation

Work, decision rights and the service model are redesigned around data.

Routine queries go to self service; HR partners move to workforce decisions; record, engagement and insight are linked

Only the third changes how value is created.

Phases from Verhoef et al. (2021); see also Vial (2019). Examples: Course framing.

VERHOEF ET AL. (2021) IN HR TERMS

The three phases, and the test for HR

Digitization

Converts analog information into digital form: a paper personnel file becomes a PDF.

Digitalization

Uses digital technology to run an existing process differently: a paper leave form becomes an app screen, with the same approvers and the same rules.

Digital transformation

In Verhoef and colleagues’ terms, a company wide change in which the core business model itself changes. In HR: the service model, decision rights, roles and data flows are redesigned.

EXAM LENS

The HR test: after the change, does anyone decide differently, faster, or with better evidence? Most “digital HR” projects stop at the first or second phase.

What each phase demands

Core claim

Three phases of digital transformation. Most of the literature holds that the first two, more incremental phases are needed to reach the third. Incumbents with legacy systems often start small and transform gradually.

Resources

Each phase makes different demands on four digital resources: digital assets, digital agility, digital networking capability, big data analytics capability.

Organization and metrics

Transformation calls for separate, agile units, and for digital KPIs such as user experience and active users, beyond cost to serve and ROI.

LIMITS

Course framing, not a finding of the paper: many firms that report a transformation have in fact digitalized. Digitalization is mainly a technology project; transformation is mainly a change project, and fails for change reasons: resistance, unclear ownership, misaligned incentives. Real organizations also sit in different phases in different functions at once.

VIAL (2019)

Eight building blocks of digital transformation

1 Digital technologies
Social, mobile, analytics, cloud, IoT (SMACIT)

→

2 Disruptions
Consumer expectations · competition · data availability

→

3 Strategic responses
Digital business strategy · digital transformation strategy

→

4 Changes in value creation paths
Value propositions · value networks · digital channels · agility and ambidexterity

5 Structural changes (enable)
Structure · culture · leadership roles · **employee roles and skills**

6 Organizational barriers (hinder)
Inertia · resistance

7 Positive impacts
Operational efficiency · organizational performance · industry and society

8 Negative impacts
Security and privacy

Redrawn from Vial (2019), Figure 1 and Tables 3 to 8. Employee roles and skills is the direct HR hook.

THEORY IN DEPTH · VIAL (2019)

What Vial adds for HR

Definition

From a review of 282 works: a process that improves an entity by triggering significant changes to its properties through combinations of information, computing, communication and connectivity technologies. The entity can be a firm, an industry or a society.

The HR hook

Transformation pushes employees into roles outside their usual functions. Supporting them through that transition is a challenge larger than the HR function alone.

Barriers

Inertia: existing systems and routines hold the old way in place. Resistance: people push back on what the change asks of them.

IN PRACTICE · COURSE FRAMING

A decade old on premise HRIS is inertia; a long serving HR officer with deep process knowledge who fears automation is the resistance the framework predicts. Reskilling and redeploying that officer is part of the transformation, not a side issue.

THE DISTINCTION THE COURSE TURNS ON

What moves when a change goes beyond digitization

Same process, on a screen	Different process, because the constraint moved
The form becomes a PDF	The step is questioned: why does it exist at all?
The register becomes a spreadsheet	The decision right moves to where the data is
The queue becomes an inbox	The service reaches people it never reached
The old rule runs automatically, including its errors	Roles are redesigned, not merely retooled
Cost falls a little; experience rarely improves	The measure of success itself changes

EXAM LENS

The left column covers digitization and much digitalization; the right column is transformation. Examiners reward answers that show which constraint moved, not answers that list technologies.

Course framing, built on Verhoef et al. (2021).

A TEST YOU CAN USE

Spotting “paper on a screen”

Q1 Did the steps change, or only the medium?

No →

Q2 Did anyone’s decision rights or speed of decision change?

No →

Q3 Is the data captured reused anywhere else?

No →

Three “no” answers: **digitization only**

A “yes” to any question means the change has gone beyond digitization; use the three phases to say how far.

Course framing, built on Verhoef et al. (2021).

APPLY THE DISTINCTION

Classify each example: which of the three phases?

1

An annual paper engagement survey is reissued as a PDF form emailed to all staff.

2

Leave applications move from a register to a shared spreadsheet that HR updates each evening.

3

Frontline staff get a vernacular mobile app that shows leave balances and approves leave instantly.

4

An applicant tracking system ranks CVs using the same rule hiring managers always used.

5

A chatbot answers payroll queries, and payroll clerks are retrained as data quality owners.

Your task

Place each in a phase, then justify the hardest one with the “paper on a screen” test.

APPLY THE DISTINCTION

Digital or transformed?

Teaching case (Course framing)

A Bhopal cement company reports its “digital HR transformation”: payslips emailed as PDFs; leave requested through an online form that HR still re keys into payroll; and a new analytics team that redesigned shift planning around data on absence and output.

Work it through

1. Classify each change as digitization, digitalization or transformation.
2. Which change alters roles and decisions?
3. What would make the leave process digitalization done well?
4. Which of Schwab’s three features does the company face?

EXAM LENS

Check your classification against the three phases and the test, not against how much technology each change uses.

APPLY: THE CASE FOR CHANGE

Making the case to a skeptical CFO

1 · Lead with cost already carried

Licenses for many systems, duplicate data entry, high attrition: costs the CFO already pays.

2 · Show what the spend buys

Current spend buys low adoption; Stone et al.’s efficiency is not effectiveness.

3 · Ask for a staged investment

With a measure the CFO can check at each stage before releasing the next.

4 · Avoid technology vocabulary

The CFO is buying outcomes, not tools.

Course framing; the Lesson Plan’s Week 1 consolidation task.

STONE, DEADRICK, LUKASZEWSKI AND JOHNSON (2015)

Efficiency is not effectiveness

Technology has transformed how HR collects, stores and uses information. Yet relatively little research shows that it helps organizations attract, select, develop, motivate and retain talented and diverse employees.

THE FOUR LIMITATIONS OF CURRENT SYSTEMS

One way

Communication runs in one direction

Impersonal

Little personal contact

Passive

Waits for the user to act

Distance

An artificial gap between people and the organization

They can also shut out people with low computer skills.

Stone et al. (2015), Human Resource Management Review, 25(2), 216 to 231. The four labels are the authors’; the short glosses are Course framing.

STONE ET AL. (2015)

How technology changes four areas of HR

Acquiring talent

Online recruitment and selection reach wider pools, with new fairness and privacy risks.

Developing people

Electronic learning shifts responsibility for learning toward employees.

Managing performance

Electronic monitoring and feedback raise questions of trust and privacy.

Delivering HR services

Self service moves work from HR staff to employees and managers.

Summarized from Stone et al. (2015).

STONE ET AL. (2015)

What technology delivers, and what the evidence questions

HR process	Efficiency: what it delivers	Effectiveness: what the evidence questions
Recruitment	More applicants	Not higher quality ones; may not improve diversity
Selection	Screening at scale	Keywords not always based on job analysis; scores can reflect computer skill
Training	Flexible and cheaper	Depends on learner control, practice, interaction, feedback; can isolate
Performance	More frequent feedback	Possibly lower rating quality; managers may rely on data and interact less
Pay and self service	Lower cost, time and errors	Effect on motivation unclear; self service shifts workload and stays one way

Summarized from Stone et al. (2015). The authors favor blended learning and interactive, two way tools as remedies.

Process by process: the detail examiners reward

Process	What the review reports
Recruitment	More applicants but not better ones; extra volume raises administrative cost; access to and comfort with technology differ across groups, so diversity may not improve
Selection	In one study, keywords were not always based on job analysis; computerized tests are not always equivalent to paper ones and can reflect computer skill and anxiety; video interviews carry less information and are viewed less positively
Training	Effectiveness depends on learner control, practice, interaction and feedback; trainees often prefer face to face, so the authors favor blended learning
Performance	Reactions are mixed; some evidence of lower rating quality; with automatic data capture, managers may rely on data and interact less
Pay and self service	Unclear that it motivates or retains, partly because diverse employees value different rewards; self service transfers HR’s workload to managers and employees and stays one way

STONE ET AL. (2015)

Two questions for the future of HR technology

Question 1

Is the real goal of eHR efficiency (lower cost, faster delivery) or effectiveness (attracting, motivating and retaining a talented, diverse workforce)?

Question 2

Does eHR push HR toward a transaction based rather than a relationship based approach to managing people?

Their position

Technology should enhance, not replace, managers and HR professionals.

IN PRACTICE · COURSE FRAMING

Self service that moves HR tasks onto frontline staff with the weakest devices is efficient for HR and costly for them. Common mistake: arguing for a system on speed or cost savings alone.

THEORY IN DEPTH · ZUBOFF (1988)

Technology can automate, or it can informate

The core idea

Automating replaces human effort. Informating produces new information about the work itself, making processes visible.

How it works

The same system does both. The choice is whether the new information empowers workers or only watches them.

Limits and critique

Written about the computerizing workplaces of the 1980s, before the web. Zuboff (2019) later argued that informating turned into surveillance for profit.

IN PRACTICE

An employee self service app automates leave requests; it also informates, because every login tells HR who uses it and who does not.

EXAM LENS

Use Zuboff to ask who benefits from the data a system creates. Pair her 1988 and 2019 books to show how the argument evolved.

THEORY IN DEPTH · AUTOR, LEVY AND MURNANE (2003)

Computers replace routine tasks, not whole jobs

The core idea

Jobs are bundles of tasks. Computers substitute for routine tasks and complement nonroutine analytic and interactive ones.

How it works

Routine cognitive tasks (record keeping) and routine manual tasks follow explicit rules, so they go first; judgment and personal interaction resist.

Limits and critique

AI now reaches some nonroutine cognitive tasks, so the routine line keeps moving (Course framing).

IN PRACTICE

An HR officer’s data entry is routine; their process knowledge and judgment are not. Redesign the job rather than delete it.

EXAM LENS

Strong answers analyze tasks, not job titles. Note how generative AI challenges the original routine and nonroutine boundary.

THEORY IN DEPTH · ORLIKOWSKI (1992)

Technology shapes people, and people reshape technology

The core idea

The duality of technology: systems are designed by people, then constrain them, and are reinterpreted in use.

How it works

Users adapt, work around, or ignore features. The system that runs is not the system that was bought.

Limits and critique

The idea explains a great deal after the fact but predicts little in advance (Course framing).

IN PRACTICE

Employees often keep private spreadsheets alongside an official HRIS: a workaround showing that the official system did not fit their work.

EXAM LENS

Use Orlikowski to explain why adoption figures mislead: high usage of a workaround can signal failure of the official system.

THE CONSEQUENCE

How the HR function’s work is restacked

Designing services
employee experience

Stewarding data
privacy and quality

Governing algorithms
fairness and accountability

Leading change
change agency (Week 11)

Processing transactions: does not disappear; it is automated underneath the new roles

EXAM LENS

Answers that say HR “becomes strategic” earn little. Name the role, the data it needs, and who it serves.

THE CONSEQUENCE

Four shifts in the HR role

From records to evidence

HR must prove effects, not report activity. “We ran 40 sessions” stops being an answer.

From process owner to product owner

HR services become interfaces people either use or abandon; adoption becomes an HR metric.

From administrator to governor

Someone must own algorithmic fairness, consent and data residency. In practice, that is HR.

From employee relations to workforce relations

Employees, contractors, apprentices and platform workers: several populations, one duty of care.

Course framing.

BHARADWAJ, EL SAWY, PAVLOU AND VENKATRAMAN (2013)

Digital business strategy

Digital strategy is not a functional IT strategy serving business strategy; digital resources now reshape business strategy itself. Any firm can buy the same technology, so advantage comes from the capability to use it (a resource based argument).

Scope

Ecosystem and functional reach: talent platforms; suites that blur HR's internal silos

Scale

Near zero marginal cost, network effects, data that improves algorithms

Speed

Real time response, rapid experiment, continuous evolution, through modular API based systems

Sources of value

Efficiency, employee experience, business model innovation such as moving from reactive hiring to talent pipelines

FOR HR

For HR: digital HR strategy cannot be set apart from business strategy; it must design for an ecosystem that includes gig workers and contractors; and speed of adaptation becomes a source of advantage.

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
Going paperless is digital transformation	It is digitization; transformation changes processes, roles and value
Industry 4.0 is about factories only	Schwab: systems impact reaches every function, including HR
Technology makes HR more effective by itself	Stone et al.: effectiveness depends on design, use and context
More systems, licenses or logins show transformation	Counting tools measures purchase, not change; apply the “paper on a screen” test
High usage proves a system works	Orlikowski: heavy use of a workaround can signal that the official system failed
Automation deletes jobs	Autor, Levy and Murnane: it changes the task mix inside jobs

COMPARE AND EVALUATE

Eight lenses on the same digital change

Theory	Question it answers	Use it to	Watch out for
Schwab (2016)	Why is this change different in kind?	Frame the drivers	Critics: overstates novelty; high bandwidth view
Verhoef et al. (2021)	How deep is the change?	Classify and sequence initiatives	Phases blur within one firm *
Vial (2019)	What does transformation set in motion?	Trace strategy, structure, barriers	A synthesis, not a tested causal model *
Stone et al. (2015)	Is HR better, not just faster?	Test a business case	Written before generative AI
Zuboff (1988; 2019)	Who gains from the data?	Analyze power and surveillance	Can underplay benefits to workers *
Autor, Levy and Murnane (2003)	Which tasks will change?	Redesign jobs, plan reskilling	The routine line keeps moving *
Orlikowski (1992)	Why does the system in use differ?	Explain adoption and workarounds	Weak at prediction *
Bharadwaj et al. (2013)	How does digital reshape strategy?	Read scope, scale, speed, value	Firm level; HR use is an application *

* Course framing: the course’s reading of the limit, not a claim made by the authors.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

Using Schwab’s account of the fourth industrial revolution and Verhoef and colleagues’ three phases, assess whether an Indian bank’s HR technology program is digital transformation or digitization. **20 marks**

A strong answer: explains velocity, breadth and depth, and systems impact; distinguishes the three phases with examples from the case; shows what would need to change for transformation; notes one risk or limit raised by Stone and colleagues.

“Most digital HR programs are digitization projects wearing a transformation budget.” Critically evaluate with reference to Verhoef et al. (2021) and Indian examples. **20 marks**

A strong answer: defines the three phases precisely; tests each example against which constraint moved; uses Orlikowski, Zuboff or Vial to explain why programs stall; ends with conditions under which the statement is false.

Industry 4.0, artificial intelligence and hybrid work are external forces, not HR initiatives. Discuss their combined implications for the HR function in India. **20 marks**

A strong answer: applies Schwab’s three criteria; links each force to specific HR processes; addresses the frontline digital divide, the DPDP Act and platform work; evaluates the equity trap rather than only describing it.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

A vendor tells an Indian hospital chain that its HR suite cut time to hire sharply and should be rolled out to all nursing staff. Evaluate the claim using Stone et al. (2015). **20 marks**

A strong answer: separates efficiency from effectiveness; asks for quality of hire, diversity and retention evidence; applies the four limitations to nurses; states what evidence would settle it.

Using Autor, Levy and Murnane (2003), explain why automation changes tasks rather than eliminating jobs. What is the main limitation of this view today? **10 marks**

A strong answer: explains the task model with an HR example, then names the moving routine line under AI as the limitation.

Using Vial (2019), explain why a digital transformation can stall even when the technology works. **10 marks**

A strong answer: names inertia and resistance as barriers; links them to employee roles and skills; gives one Indian example; notes one limit of the framework.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Three external forces, not three HR projects.**
   Industry 4.0, AI and hybrid work arrived together and landed on HR; use all three of Schwab’s criteria.

2. **Digitization moves a process onto a screen; transformation redesigns work, decisions and service.**
   Classify with Verhoef; explain stalls with Vial and Orlikowski.

3. **Efficiency is not effectiveness.**
   With Stone et al. and Zuboff, ask what got better, for whom, and who holds the data.

4. **Reach is the constraint, not features.**
   A design that works only for the desk based, English fluent minority divides the workforce rather than transforming it.

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- Schwab, K. (2016). *The fourth industrial revolution*, Introduction and Part I. World Economic Forum.
- Stone, D. L., Deadrick, D. L., Lukaszewski, K. M., & Johnson, R. (2015). The influence of technology on the future of human resource management. *Human Resource Management Review, 25*(2), 216 to 231.

NEXT WEEK

- Westerman, Bonnet and McAfee (2014), *Leading Digital*, Chapters 1 and 2.
- Scully-Russ and Torraco (2020), pp. 73 to 77.
- Deloitte, Global Human Capital Trends (latest), selected sections.

> **NOTE**
> The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics, 118*(4), 1279 to 1333. https://doi.org/10.1162/003355303322552801
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- Gratton, L. (2021). How to do hybrid right. *Harvard Business Review, 99*(3), 66 to 74.
- Kagermann, H., Wahlster, W., & Helbig, J. (2013). *Recommendations for implementing the strategic initiative INDUSTRIE 4.0*. acatech.
- NITI Aayog. (2022). *India's booming gig and platform economy: Perspectives and recommendations on the future of work*. Government of India.
- Orlikowski, W. J. (1992). The duality of technology: Rethinking the concept of technology in organizations. *Organization Science, 3*(3), 398 to 427. https://doi.org/10.1287/orsc.3.3.398
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